

From fibrosis to regeneration: cardiac fibroblasts in adult acomys cahirinus

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Cardiovascular diseases (CVD) remain the leading cause of death worldwide, with myocardial infarction (MI) as a main contributor (1, 2). Mortality stems from the growing prevalence of cardiovascular risk factors and poor regenerative capacity of the human heart (3, 4).

Regenerative models such as zebrafish or neonatal rodents have provided valuable insights into cardiac regeneration, largely due to their ability to induce cardiomyocyte (CM) proliferation, something rarely achievable in the adult human heart (5). Recently, attention has turned to non-CM populations, particularly cardiac fibroblasts (CF), which play essential roles in tissue remodelling and extracellular matrix (ECM) production. Fibrosis, driven by CF activation and differentiation into myofibroblasts, is a major barrier to regeneration (6, 7). In these animal models, fibrosis is overcome, but the dominant role of CM proliferation complicates efforts to isolate CF's contribution.

The African spiny mouse (*Acomys cahirinus*), an adult mammal capable of scarless healing across multiple tissues, has emerged as a promising model to bridge this gap. In the heart, *Acomys* displays enhanced healing after MI, with reduced fibrosis and preserved function. Although some CM proliferation may occur, the dominant feature is a loose, afibrotic ECM (8-10). We hypothesize that CF are central mediators of the regenerative response observed in *Acomys* hearts.

To test this, we isolated and characterized CF from *Acomys* and *Mus*, analysing their responses to pro-fibrotic biochemical (TGF β) and mechanical stimuli. *Acomys* CF showed enhanced proliferation and migration, yet a blunted response to TGF β and substrate stiffness (Fig1), with limited α SMA induction, reduced YAP nuclear translocation and less matix engagement (focal adhesions). In 3D spheroids, they produced a loose, fibronectin-rich ECM with low collagen deposition, in contrast to the dense collagen matrix formed by *Mus* CF (Fig2).

Over time, *Acomys* CF showed increased matrix remodelling, particularly of collagen, while maintaining a fibronectin-rich ECM distinct from persistent thick collagen bundles produced by *Mus* CF. To test reversibility, we withdrew TGF β in 3D cultures. *Acomys* spheroids responded with morphological changes, decreased ECM deposition, and partial loss of activation markers, supporting a dynamic and reversible fibrotic program. Moreover, *Acomys* CF exhibited increased apoptotic activity compared to *Mus*, particularly under TGF β stimulation, suggesting a mechanism to prevent sustained fibrotic activation.

These findings reveal differences in how *Acomys* and *Mus* CF respond to profibrotic stimuli, establishing a framework to: (i) better understand CF contributions to regeneration and how fibrotic traits like proliferation or migration affect outcomes; (ii) guide the development of regenerative therapies focused on ECM modulation; and (iii) gain insights into *Acomys*, with potential applications beyond cardiac regeneration.

Figure 1

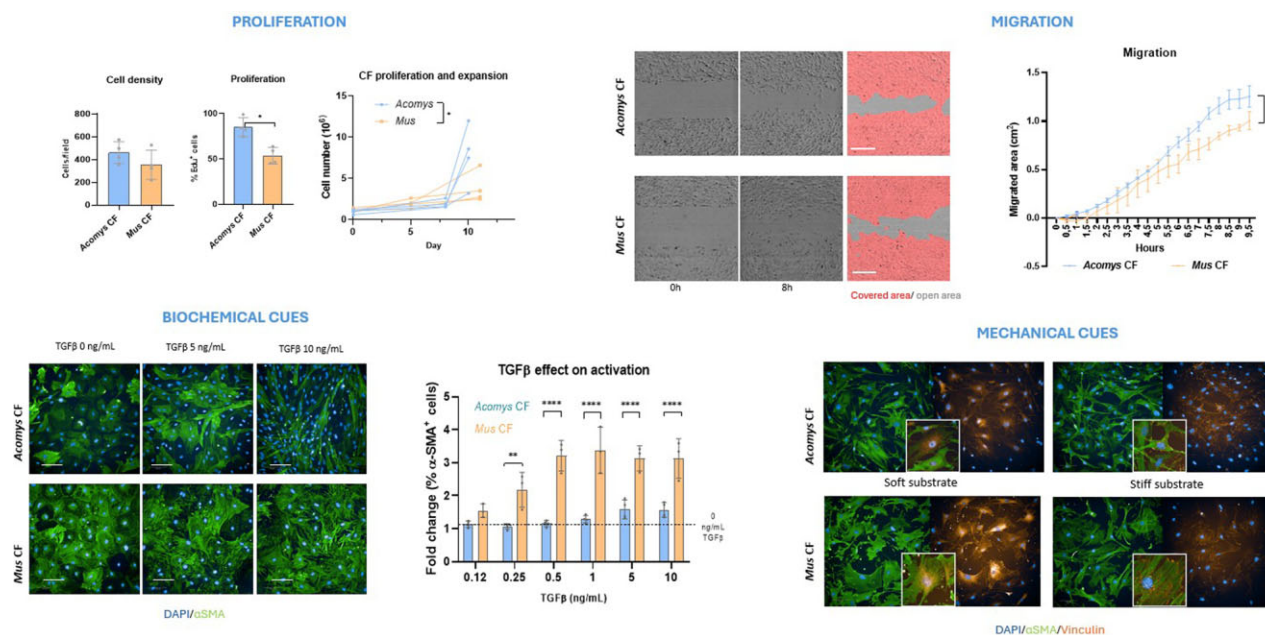


Figure 2

